
Is This 3D Asset Usable? Beyond the Beauty Render for Asset Curation

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Abstract

1 Large 3D repositories make it easy to collect assets, but difficult to determine
2 whether they can be reused reliably across scene assembly, editing, rendering, game
3 production, and training-data construction. An asset can look plausible in RGB
4 while hiding disconnected geometry, problematic UV organization, or materials
5 that fail to resolve correctly after import. We introduce *Diagnostic Rendered*
6 *Evidence (DRE)*: aligned beauty, wireframe, and UV-seam-plus-wireframe views
7 scored by pass-specific frozen ViT classifiers and combined through late fusion. On
8 1,234 human labeled TexVerse assets, nested grouped evaluation achieves 86.36%
9 balanced accuracy and 88.32% bad-asset recall. On a separate, balanced 764
10 external reference test, DRE achieved 88.9% balanced reference-label accuracy.
11 Our method requires only an imported asset and renderer-derived evidence, without
12 repository identity, creator, popularity, text, or source metadata at inference. Our
13 results support cross-corpus portability and show that purpose built diagnostic
14 views expose defects that just the beauty views can miss.

15 1 Introduction

16 Consider a common workflow where we download a model from a large 3D repository. It can
17 look perfectly fine and still be difficult to use. A few RGB views look convincing: the shape is
18 recognizable, the texture appears detailed, and nothing is obviously broken. Yet once the asset enters
19 a 3D workflow, different problems can emerge. The mesh may contain disconnected or irregular
20 geometry, the UV layout may complicate texturing, or materials may fail to resolve correctly after
21 import. None of these problems necessarily prevents the asset from producing an attractive preview.

22 Repository assets are increasingly reused as structured 3D objects: retrieved and arranged into scenes
23 [25, 1, 17], edited [11], rigged for animation [15], or used in modern image-to-3D pipelines that
24 explicitly represent geometry, UVs, textures, and PBR attributes [3, 19, 23]. Across these uses,
25 plausible RGB appearance does not guarantee reliable reuse.

26 Large repositories such as ShapeNet [4], Objaverse and Objaverse-XL [8, 7], and TexVerse [26] make
27 this a repository-scale curation problem: *How can we determine whether a 3D asset is fit for reuse*
28 *before selecting it from a large repository?*

29 This requires expert review, that considers appearance, geometry, UV structure, and material behav-
30 ior. We therefore construct a 1,234-asset TexVerse set annotated by 3D researchers and introduce
31 **diagnostic rendered evidence**: aligned beauty, wireframe, and UV-seam-plus-wireframe views that
32 expose complementary properties of the same asset. Separate models score each evidence type before
33 late fusion produces the final keep/reject decision. Our scope is *operational fitness* for reusable
34 textured/PBR assets rather than universal aesthetic quality or asset provenance. On our human-labeled
35 assets, our method reaches 86.36% balanced accuracy and 88.32% bad-asset recall. Controlled abla-
36 tions show that the structural diagnostic passes provide information beyond a matched beauty-only

37 representation. Our contributions are: 1: We introduce *diagnostic rendered evidence* for 3D asset
 38 curation using aligned beauty, wireframe, and UV-seam-plus-wireframe views. 2: We develop a
 39 robust rendering pipeline for heterogeneous repository assets, including guarded material fallbacks
 40 and explicit material-resolution provenance. 3: We provide a 1,234-asset expert-annotated benchmark
 41 for reusable textured/PBR assets, with a keep/reject rubric covering geometry, UV, material, render,
 42 and composition failures.

43 2 Related Work and Scope

44 **3D assets as reusable components:** Recent systems increasingly treat existing 3D assets as reusable
 45 components. Holodeck retrieves and arranges Objaverse assets for scene generation [25]; Open-
 46 Universe similarly retrieves meshes from uncuration databases and optimizes their placement [1]; and
 47 Lay-A-Scene studies object arrangement [17]. Existing assets are also edited [11] and automatically
 48 rigged for animation [15]. These settings motivate evaluating asset properties that are not fully
 49 captured by a beauty render.

50 **3D curation and quality assessment:** Objaverse++ and MeshFleet use filtering and human review
 51 to construct higher-quality 3D corpora [14, 2], while generative-3D benchmarks evaluate geometry,
 52 texture, and cross-view consistency [13, 20, 27, 10]. Perceptual mesh-quality studies likewise show
 53 that geometry and texture jointly affect human judgments [12, 16]. Full-reference metrics assume a
 54 known reference [16, 24]; repository curation instead requires judging a single asset without one.

55 **Structured evidence for asset quality:** Modern 3D systems explicitly represent geometry, UVs, and
 56 materials. Hunyuan3D separates mesh and texture generation [28], SF3D predicts UV-unwrapped
 57 geometry and material properties [3], and AssetGen reconstructs geometry and PBR appearance
 58 [19]. Related work further models UV and material structure explicitly [5, 22, 23], while UV-specific
 59 methods motivate interpreting parameterization in surface context [21, 18]. Our method uses rendered
 60 views of these structures as no-reference evidence for asset curation.

61 3 Methodology

62 **Evidence Generation and Provenance:** For each imported GLB, we render synchronized diagnostic
 63 passes using fixed camera poses, lighting, background, and framing. Four aligned views are generated
 64 per pass and scored independently before late fusion. The *beauty* pass captures overall appearance and
 65 visible rendering or material failures, but may hide structural defects through texture, shading, and
 66 lighting. The *wireframe* pass exposes mesh structure, including disconnected components, irregular
 67 triangulation, excessive density, and geometric clutter. These patterns are treated as evidence rather
 68 than fixed quality rules. The *UV-seam-plus-wireframe* pass overlays UV boundaries on the mesh,
 69 allowing seam organization to be interpreted in its 3D surface context rather than from a flattened
 70 atlas alone.

71 **Repository-Agnostic Evidence:** DRE is designed around information that can be obtained from
 72 an imported asset rather than from a particular repository. The renderer uses no asset ID, creator,
 73 source label, caption, popularity statistic, or repository-specific metadata. It first resolves standard
 74 Principled-BSDF base-color inputs, then follows supported graph-connected image textures and
 75 guarded fallbacks; when resolution is not possible, it records an explicit unresolved-material state
 76 rather than inventing a material interpretation. We also generated 2D UV-island layouts and mea-
 77 surements such as coverage, angular distortion, and texel density. These signals were reviewed by
 78 3D experts and evaluated in additional classifier configurations, but provided little improvement
 79 over the surface-aligned diagnostic views. We therefore retain them for review and audit rather
 80 than classifier input, particularly since UV packing conventions vary across asset categories and
 81 authoring workflows. Figure 1 illustrates the synchronized evidence available to both the reviewer
 82 and the classifier. No individual diagnostic pass is treated as ground truth. Human review separates
 83 several kinds of concern: low overall curation value, geometry integrity or clutter, UV organization,
 84 material-resolution anomalies, missing evidence, and other import or composition problems.

85 **Multi-view Scoring and Fusion:** For pass m and view v , a frozen ImageNet ViT-B/16 encoder
 86 [9] produces $z_{m,v} \in \mathbb{R}^{768}$. We summarize the four aligned views with their feature-wise mean and
 87 standard deviation. A standardized, class-balanced logistic-regression model is then trained separately

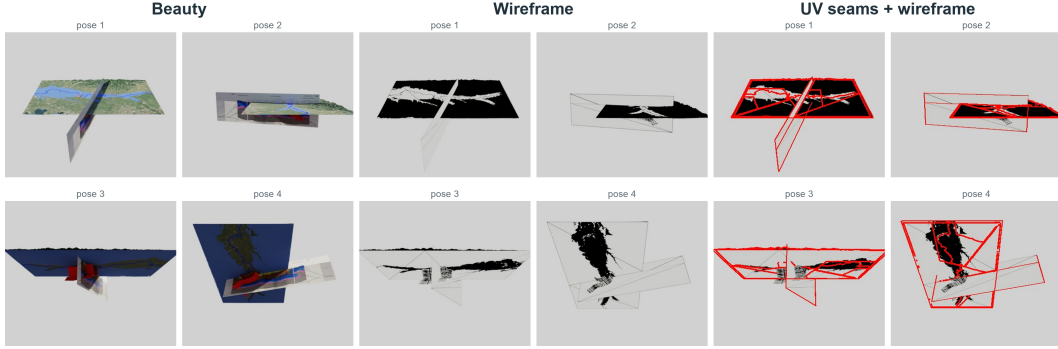


Figure 1: Aligned beauty, wireframe, and UV-seam-plus-wireframe evidence. The external baseline assigns this asset *Superior*, but the diagnostic views expose its disconnected planar fragments.

(a) Selected fusion		(b) Representation-controlled evidence ablation			
Metric	Value	Evidence configuration	Acc.	BA	BR
Accuracy	85.58%	Beauty	73.26	71.26	66.30
Balanced accuracy	86.36%	Wireframe	81.20	80.35	78.26
Bad-asset recall	88.32%	UV seams + wire	84.60	84.97	85.87
Good-asset recall	84.41%	Beauty + wireframe	81.28	80.65	79.08
Good-asset precision	94.44%	Beauty + UV seams + wire	84.28	84.03	83.42
Bad predicted good	43	Wireframe + UV seams + wire	84.04	84.88	86.96
Good predicted bad	135	All three passes	82.90	84.15	87.23

Table 1: Grouped development results on 1,234 human-labeled assets. (a) Selected nested grouped fusion. (b) Ablations holding labels, groups, ViT representation, scoring family, and threshold-selection protocol fixed. Acc.: accuracy, BA: balanced accuracy, BR: bad-asset recall.

for beauty, wireframe, and UV-seam-plus-wireframe evidence. Each model produces a probability $p_m(\text{good})$ describing how strongly that particular type of evidence supports keeping the asset.

4 Learning and Evaluation

We use TexVerse [26] to create our set under CC BY, CC BY-SA, or CC0 licenses. Our supervised set contains 1,234 assets annotated by expert 3D researchers: 866 good and 368 bad. An additional 6,685 assets based on the deterministic render QA and their metadata are used as less weighted auxiliary training data. We use five-fold nested stratified group cross-validation, grouping by creator or source where available. Inner folds train the pass-specific classifiers, generate out-of-fold predictions for fusion, and select weak-label weights, regularization, pass combinations, and the decision threshold. Auxiliary samples sharing a group with the held-out fold are excluded. Metadata and strict full-PBR inventory provide weak-label sources, while terms such as *Rodin*, *photogrammetry*, or *scan* identify low-weight negative candidates rather than evaluation labels. Face count, texture resolution, creator, and source are retained only for audit and stratification. Table 1 reports outer-fold predictions.

Modality Ablation: Table 1 keeps the labels, grouped folds, ViT representation, classifier family, fusion method, and threshold-selection protocol fixed while changing only the render evidence. UV-seam-wire achieves the highest balanced accuracy, while the full three-pass model gives the highest bad recall and fewest false accepts. Thus, the benefit is not simply due to using more views: the structural passes provide information complementary to beauty renders. We exclude material-only views because their appearance is difficult to separate from renderer behavior. A gray render, for example, may indicate an intentional material, scalar input, vertex color, missing texture, or failed graph resolution. Material behavior is therefore audited separately rather than used as a learned visual modality.

Baselines: The matched beauty-only ViT in Table 1 is our primary representation baseline. It keeps the labels, groups, backbone, classifier, fusion, and threshold-selection protocol fixed, isolating the contribution of diagnostic evidence. We also evaluate the released Objaverse++ [14] on our dataset using its 40-view protocol. Our method uses no popularity, creator, source, or text features at

Table 2: Objaverse++ versus DRE (ours) on expert-labeled TexVerse assets and a cross-source reference set.

Set	Model	Acc./BA $\uparrow$	G-Rec. $\uparrow$	B-Rec. $\uparrow$	BAR $\downarrow$
TexVerse (1,234)	Objaverse++	44.65 / 55.72	28.29	83.15	16.85
	DRE (ours)	85.58 / 86.36	84.41	88.32	11.68
Cross-source (764)	Objaverse++	46.99 / 46.99	88.74	5.24	94.76
	DRE (ours)	88.87 / 88.87	80.37	97.38	2.62

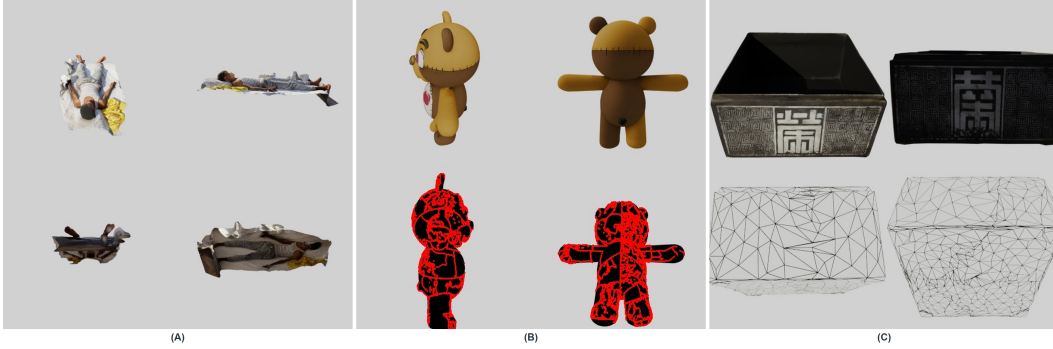


Figure 2: Three failure cases revealed by diagnostic rendered evidence. (A) An incomplete scan-like fragment is visibly unusable in beauty views. (B) A plausible-looking character shows unusually broad UV-seam coverage. (C) A regression failure rated *Superior* by the external baseline reveals disconnected and intersecting geometry in wireframe views. These examples illustrate diagnostic evidence rather than universal rejection rules.

inference and requires 12 diagnostic images per asset, compared with 40 RGB views for Objaverse++. Objaverse++ outputs are mapped as High/Superior = keep and Low/Medium = reject.

Our method retains 731 of 866 good assets and admits 43 of 368 bad assets. Objaverse++ retains 245 good assets and admits 62 bad assets. A paired exact McNemar test gives $p = 1.31 \times 10^{-102}$, with 568 assets correct only for our method and 63 correct only for Objaverse++. Further we additionally evaluate cross-corpus portability on a fixed, balanced external reference set of 382 good and 382 bad assets from strict-PBR ABO [6] and Objaverse assets. DRE achieves 88.9% balanced reference-label accuracy, with 80.4% good-reference recall and 97.4% bad-reference recall. Objaverse got about 47.0% balanced reference-label accuracy: although it retains 88.7% of the strict-PBR reference assets, it rejects only 5.2% of the bad assets, admitting 94.8% as good.

Figure 2 shows how diagnostic passes make different failure modes inspectable. Some failures are visible in conventional appearance renders, while others become clear only when the same camera poses expose geometry or UV seams. The diagnostic representation does not turn mesh density or seam count into hard rules. It gives reviewers and the classifier evidence that can be interpreted in the context of the asset and its intended downstream use.

5 Conclusion

A beauty render can indicate whether an asset looks plausible, but not whether it is operationally fit for reliable reuse. Diagnostic evidence for appearance, topology, UV structure, and material resolution makes the relevant conditions inspectable from the asset itself. By separating these evidence types before fusion, DRE provides a repository-agnostic basis for quality-controlled asset selection for various workflows.

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